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ENGINEERED WOVEN FABRIC AND ARTICLE OF CLOTHING INCLUDING THE WOVEN FABRIC

Abstract

Examples of an article of clothing including an engineered woven fabric are described. The article of clothing includes a single-layer woven portion and a double-layer woven portion seamlessly joined with the single-layer woven portion to form a pocket. The pocket includes closed side edges seamlessly joined with the single-layer woven portion and an opening. The single-layer woven portion is engineered with a first stretching modulus zone with a first level of stretchability and a second stretching modulus zone with a second level of stretchability, such that the first level of stretchability is greater than the second level of stretchability. The second stretching modulus zone is located around the closed edges of the pocket to provide support around the pocket.

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Background/Summary

BACKGROUND

[0001] The present embodiments relate to garments and more particularly to garments including an engineered woven fabric with pocket bags seamlessly woven therein.

[0002] Typically, pocket bags in garments are designed and constructed by cutting a fabric panel and sewing it to the garment, such that extra fabric is needed to form the pocket bag. The process of cutting and sewing is time consuming, increases the fabric waste, and the resulting pocket bags may be unattractive to users of the garment as they may appear baggy, hanging, and distracting.

[0003] There exists a need in the art for a garment formed with reduced sewing while still including double layered structures such as pocket bags.

SUMMARY

[0004] In one aspect, the disclosure provides an article of clothing including an engineered woven fabric. The engineered woven fabric includes a single-layer woven portion having at least one warp yarn and at least one weft yarn. The single-layer woven portion engineered with a first stretching modulus zone with a first level of stretchability and a second stretching modulus zone with a second level of stretchability. The first level of stretchability being greater than the second level of stretchability. The engineered woven fabric also includes a double-layer woven portion seamlessly joined with the single layer woven portion and having two independent woven layers each having at least one warp yarn and at least one weft yarn. The double-layer woven portion forming a pocket with closed edges seamlessly joined with the single-layer woven portion and an opening. The single-layer woven portion is engineered such that the second stretching modulus zone is disposed around the closed edges of the pocket to provide support around the pocket.

[0005] In another aspect, the disclosure provides an article of clothing including an engineered woven fabric. The engineered woven fabric includes a pocket formed in the engineered woven fabric. The pocket having a plurality of closed edges disposed around an outer perimeter of the pocket and a free edge that provides an opening into a cavity of the pocket. The engineered woven fabric also includes a surrounding portion of the engineered woven fabric surrounding at least two closed edges of the plurality of closed edges around the outer perimeter of the pocket. The engineered woven fabric further includes at least one additional portion of the engineered woven fabric surrounding the surrounding portion. The at least one additional portion of the engineered woven fabric includes a first stretching modulus zone having a first level of stretchability and the surrounding portion includes a second stretching modulus zone having a second level of stretchability. The second level of stretchability being less than the first level of stretchability to provide support around the pocket.

[0006] In another aspect, the disclosure provides a method of forming an article of clothing including an engineered woven fabric having a seamless pocket. The method includes weaving an engineered woven fabric with: (1) a single-layer woven portion having at least one warp yarn and

at least one weft yarn, and (2) a double-layer woven portion having two independent woven layers each having at least one warp yarn and at least one weft yarn. The double-layer woven portion is seamlessly joined with the single layer woven portion during the weaving process. The method of weaving the single-layer woven portion further includes weaving a first stretching modulus zone in a first portion of the engineered woven fabric with a first level of stretchability and weaving a second stretching modulus zone in a second portion of the engineered woven fabric with a second level of stretchability. The first level of stretchability being greater than the second level of stretchability. The method of weaving the double-layer woven portion further includes forming a pocket in the engineered woven fabric with closed edges extending around an outer perimeter of the pocket. The closed edges being seamlessly joined with the single-layer woven portion. The method further includes cutting one woven layer of the two independent woven layers along one edge of the closed edges extending around the outer perimeter of the pocket to detach the one woven layer and form a free edge. The free edge provides an opening into a cavity formed between the two independent woven layers of the pocket.
[0007] Other systems, methods, features and advantages of the disclosure will be, or will become, apparent to one of ordinary skill in the art upon examination of the following figures and detailed description. It is intended that all such additional systems, methods, features and advantages be included within this description and this summary, be within the scope of the disclosure, and be protected by the following claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] The disclosure can be better understood with reference to the following drawings and description. Throughout the drawings, reference numbers may be re-used to indicate correspondence between referenced elements. The drawings are provided to illustrate example embodiments described herein and are not intended to limit the scope of the disclosure. Sizes and relative positions of elements in the drawings are not necessarily drawn to scale. For example, the shapes and sizes of various elements and angles are not drawn to scale, and some of these elements are arbitrarily enlarged and positioned to improve drawing legibility.
[0009] FIG. **1** is a schematic view of an example embodiment of a portion of an article of clothing including an engineered woven fabric with pockets;
[0010] FIG. **2** is an enlarged schematic view of the example embodiment of a pocket seamlessly formed in the engineered woven fabric of the article of clothing;
[0011] FIG. **3** is a schematic cross-sectional view of the pocket seamlessly formed in the engineered woven fabric taken along the line shown in FIG. **2**;
[0012] FIG. **4** is a schematic view of an example embodiment of a portion of an article of clothing including an engineered woven fabric with pockets and having different stretching modulus zones;
[0013] FIG. **5** is an enlarged schematic view of an example embodiment of a pocket formed in the engineered woven fabric surrounded by different stretching modulus zones;
[0014] FIG. **6** is an enlarged representative view of a process for seamlessly forming an opening in a pocket in an engineered woven fabric;
[0015] FIGS. **7**A-**7**C illustrate an example embodiment of a process for seamlessly forming an opening in the pocket in the engineered woven fabric;
[0016] FIG. **8**A is an alternate embodiment of an arrangement for a pocket formed in the engineered woven fabric;
[0017] FIG. **8**B is another alternate embodiment of an arrangement for a pocket formed in the engineered woven fabric;
[0018] FIG. **8**C is another alternate embodiment of an arrangement for a pocket formed in the

engineered woven fabric; and
[0019] FIGS. **9**A-**9**E illustrate example embodiments of articles of clothing including an engineered woven fabric with pockets.
DETAILED DESCRIPTION
[0020] Articles of clothing having pockets seamlessly woven into an engineered woven fabric are described herein. The techniques of the present embodiments provide an engineered woven fabric and an article of clothing made from the engineered woven fabric. In various embodiments, articles of clothing with pockets or pocket bags seamlessly integrated into the article are provided, thereby minimizing fabric wastage and visual distraction.
[0021] In the following description, details are set forth to provide an understanding of the application. In some instances, certain structures, techniques, and methods have not been described or shown in detail in order not to obscure the application. In the context of the present disclosure, various terms are used in accordance with what is understood to be the ordinary meaning of those terms.
[0022] For consistency and convenience, directional adjectives are employed throughout this detailed description corresponding to the illustrated embodiments. It will be understood that each of these directional adjectives may be applied to garments or articles of clothing, as well as individual components of the article of clothing or garment. Directional terms such as “top”, “bottom”, “front”, “back”, “upper”, “lower”, “outer” and “inner” are used in the following description for the purpose of providing relative reference only, and are not intended to suggest any limitations on how any article or garment is to be positioned during use, or to be mounted in an assembly or relative to an environment. The use of the word “a” or “an” when used herein in conjunction with the term “comprising” may mean “one”, but it is also consistent with the meaning of “one or more”, “at least one” and “one or more than one”. Any element expressed in the singular form also encompasses its plural form. Any element expressed in the plural form also encompasses its singular form. The term “plurality” as used herein means more than one, for example, two or more, three or more, four or more, and the like.
[0023] Articles of clothing made from woven fabrics and methods of making such articles are described herein. A woven fabric or textile is made by interlacing warp and weft yarns using a weaving machine or loom. Warp yarns extend vertically through the fabric and weft yarns extend horizontally through the fabric. In the context of the present disclosure, the term “seamless” defines a fabric surface that is smooth and without seams or obvious joins between one part or section and the next. Thus, the parts or sections of a seamless woven fabric are formed in the same weaving process and are not separate pieces that are joined together after each is woven.
[0024] FIG. **1** illustrates an example embodiment of an article of clothing **100** (also referred to herein as “article **100**”). For clarity, the following detailed description discusses an example embodiment of a garment or article of clothing in the form of a vest, but it should be noted that the techniques described herein could be applied to any form of an article of clothing or garment including, but not limited to: jackets, coats, shirts, pants, shorts, skirts, dresses, footwear as well as other kinds of garments or clothing.
[0025] Referring now to FIG. **1**, a front side **102** of article **100** is shown. In some embodiments, article **100** may have a first side portion **104** associated with a first lateral side **10** and a second side portion **106** associated with a second lateral side **20**. In an example embodiment, first lateral side **10** corresponds to a right side of article **100** and second lateral side **20** corresponds to an opposite left side of article **100** when article **100** is worn by a wearer. In this embodiment, first side portion **104** and second side portion **106** are separated by a fastener in the form of a zipper **108** extending vertically down front side **102** of article **100** from a collar to a bottom hem.
[0026] In accordance with the techniques of the present embodiments, article **100** may be provided with one or more pockets that are seamlessly integrated into the woven fabric forming article **100**. In an example embodiment, article **100** may include a first pocket **110** seamlessly formed into the

woven fabric of first side portion **104** on first lateral side **10** and a second pocket **112** seamlessly formed into the woven fabric of second side portion **106** on second lateral side **20**. In some embodiments, pockets **110**, **112** provided on article **100** may be in the form of a pouch or pocket bag that is configured as a cavity or void space to hold or carry items or objects within the cavity or void space of the pocket. In other embodiments, pockets **110**, **112** may be in the form of a cavity or void space within article **100** to accommodate hands of a wearer of article. In still other embodiments, pockets **110**, **112** may accommodate objects and/or hands of a wearer of article **100**.

[0027] In an example embodiment, first pocket **110** and second pocket **112** are arranged on article **100** with openings facing towards the lateral sides of article **100**. As shown in FIG. **1**, opening **114** of first pocket **110** is facing towards first lateral side **10** and opening **116** of second pocket **112** is facing towards second lateral side **20**. In this embodiment, openings **114**, **116** are configured to accommodate hands of a wearer of article **100** and may also be used as a pouch or pocket bag to store items or objects. For example, in one embodiment, pockets **110**, **112** may be sized and dimensioned so as to accept items or objects such as a mobile phone, wallet, keys, sunglasses, and other items or objects of a similar size. In other embodiments, pockets **110**, **112** or additional pockets on article **100** may have larger or smaller sizes and dimensions to accept items or objects of other sizes.

[0028] In some embodiments, one or more of the pockets on article **100**, including first pocket **110** and/or second pocket **112**, may have a closure system that allows a wearer of article **100** to close and secure the opening of the pocket so that any items or objects within are secured held inside the pocket. In this embodiment, first pocket **110** has a first closure system **118** associated with opening **114** and second pocket **112** has a second closure system **120** associated with opening **116**. In one embodiment, first closure system **118** and second closure system **120** may be in the form of a zipper that may be moved back and forth to open and close each opening **114**, **116** of pockets **110**, **112**.

[0029] In other embodiments, first closure system **118** and/or second closure system **120** may be other types of fasteners, including, but not limited to hook and loop fasteners, buttons, snap fasteners, magnets, or other mechanisms that may be used to open and close openings **114**, **116**. In still other embodiments, pockets **110**, **112** may not include any closure system for openings **114**, **116**. For ease of illustration, first closure system **118** and second closure system **120** are omitted from the remaining Figures.

[0030] Referring now to FIG. **2**, an enlarged schematic view of the example embodiment of first pocket **110** is shown seamlessly formed in the engineered woven fabric of article of clothing **100**. In some embodiments, one or more pockets formed in the engineered woven fabric of article of clothing **100** may have a generally rectangular shape. In an example embodiment, first pocket **110** has an outer perimeter **200** defined by four edges, including a first edge **202**, a second edge **204**, a third edge **206**, and a fourth edge **208**. In this embodiment, outer perimeter **200** of first pocket **110** has a generally rectangular or square shape. In other embodiments, one or more pockets of article **100**, including first pocket **110** and/or second pocket **112**, may have different shapes, including but not limited to shapes shown in FIGS. **8**A-**8**C, as well as other shapes.

[0031] In accordance with the techniques described herein, in some embodiments, pockets are seamlessly formed in the engineered woven fabric of article **100** by weaving a portion of the woven fabric with a two-layer or double-layer construction. That is, while other portions of article **100** may be formed from a single layer of woven fabric, in the one or more portions of article **100** corresponding to the locations of pockets, including first pocket **110** and/or second pocket **112**, the woven fabric has a double-layer construction. With this arrangement, a cavity or void space located between the two layers of woven fabric forms the pocket (e.g., first pocket **110** and/or second pocket **112**).

[0032] As shown in FIG. **2**, first pocket **110** includes opening **114** located along fourth edge **208** of outer perimeter **200**. In an example embodiment, opening **114** is formed by detaching a portion of the two layers forming first pocket **110** so that one layer remains connected along the entirety of outer perimeter **200** and another layer is detached from the second layer along one edge of outer perimeter **200** (e.g., fourth edge **208**) to produce a free edge **210**. In this embodiment, free edge **210** is generally parallel to fourth edge **208** and is detached from the woven fabric forming article **100** to provide access through opening **114** to the cavity or void space of first pocket **110** located between the two layers of woven fabric.

[0033] FIG. **3** is a schematic cross-sectional view of first pocket **110** seamlessly formed in the engineered woven fabric taken along the line shown in FIG. **2**. In an example embodiment, first pocket **110** is seamlessly formed in the engineered woven fabric forming article **100** by weaving a double-layer woven portion having two independent woven layers each having at least one warp yarn and at least one weft yarn. The double-layer woven portion forms a pocket (e.g., first pocket **110**) having an opening along one edge (e.g., opening **114**) and closed edges around the remaining outer perimeter that are seamlessly joined with a single-layer woven portion of the engineered woven fabric.

[0034] As shown in FIG. **3**, first pocket **110** includes a cavity or void space **300** that is located between the double-layer woven portion of the engineered woven fabric formed by a first woven layer **302** and a second woven layer **304**. In this embodiment, first woven layer **302** is located on one side of cavity **300** and second woven layer **304** is located on the opposite side of cavity **300**. In some embodiments, the double-layer woven portion that includes first woven layer **302** and second woven layer **304** includes two or more warp yarns and two or more weft yarns.

[0035] In an example embodiment, one of first woven layer **302** and second woven layer **304** includes a free edge that is separated from the remaining portion of the engineered woven fabric to provide opening **114** that allows access to cavity or void space **300** of first pocket **110**. That is, while the remaining outer perimeter of first pocket **110** includes closed edges that are seamlessly joined with the surrounding single-layer woven portion, one edge of first pocket **110** is not closed (i.e., is free or open) so that it remains separated or independent from the surrounding single-layer woven portion.

[0036] In this embodiment, free edge **210** is disposed at one end of first woven layer **302** to provide opening **114** that allows access to cavity or void space **300** of first pocket **110**. For example, free edge **210** may be moved outwards away from second woven layer **304** to provide opening **114** permitting entry of a hand of a wearer of article **100**, as well as items or objects, into cavity **300** of first pocket **110**. At an opposite end **306** of first woven layer **302** from free edge **210**, first woven layer **302** and second woven layer **304** are seamlessly joined with a single-layer woven portion **308**.

[0037] In some embodiments, an article of clothing may be formed from an engineered woven fabric having different stretching modulus zones in portions of the fabric to provide varying amounts of stretch resistance and support to the article of clothing. In accordance with the techniques of the example embodiments described herein, an article of clothing may be made from an engineered woven fabric having one or more pockets seamlessly integrated into the woven fabric and the pockets may be surrounded by one or more different stretching modulus zones in portions of the woven fabric. With this arrangement, the article may be provided with a smooth visual appearance and support may also be provided to reinforce the pocket.

[0038] Referring now to FIG. **4**, an example embodiment of a portion of an article of clothing **400** including an engineered woven fabric with pockets and having different stretching modulus zones is shown. In some embodiments, a front side **402** of article **400** may have a first side portion **404** associated with first lateral side **10** and a second side portion **406** associated with second lateral side **20**. In this embodiment, first side portion **404** and second side portion **406** are separated by a fastener in the form of a zipper **408** extending vertically down front side **102** of article **100** from a collar **410** to a bottom hem **412**.

[0039] In an example embodiment, article **400** may include first pocket **110** seamless formed into the woven fabric of first side portion **404** on first lateral side **10** and second pocket **112** seamlessly formed into the woven fabric of second side portion **406** on second lateral side **20**. As described above, first pocket **110** and second pocket **112** are seamlessly formed in the engineered woven fabric of article **400** by weaving a double-layer woven portion having two independent woven layers having an opening along one edge (e.g., opening **114** of first pocket **110** and opening **116** of second pocket **112**) and closed edges around the remaining outer perimeter that are seamlessly joined with a single-layer woven portion of the engineered woven fabric of article **400**.

[0040] In some embodiments, the single-layer woven portion of the engineered woven fabric of article **400** may be provided with various stretching modulus zones that are formed through the weaving process with different amounts or degrees of stretch (i.e., stretchability). In some cases, the different stretching modulus zones may be formed using different weaving patterns or techniques (e.g., plain weave, pile weave, twill weave, etc.) or different weaving style (e.g., tight weave or loose weave) to provide different amounts or degrees of stretch. For example, some portions of the engineered woven fabric of article **400** can be tightly woven (smaller space between the threads) and other portions can be less tightly woven (bigger space between threads). In some embodiments, various stretching modulus zones can be formed using different combinations of weaving patterns/techniques and weaving styles. In other cases, the different stretching modulus zones may be formed using different yarns formed of materials having different amounts or degrees of stretch. In still other cases, a combination of different weaving patterns or techniques and yarns or different materials may be used to provide the different stretching modulus zones.

[0041] In an example embodiment, a majority of front side **402** of article **400** may be formed by a single-layer woven portion of the engineered woven fabric, including parts or sections of first side portion **404** and second side portion **406**. On first side portion **404**, a first upper portion **414** extends from collar **410** in a downward direction towards bottom hem **412**. First upper portion **414** is associated with a stretching modulus zone **418** having a level of stretchability. First side portion **404** also includes a first lower portion **416** that extends in an upwards direction from bottom hem **412** towards collar **410**. First lower portion **416** is associated with a stretching modulus zone **420** having a level of stretchability. In some embodiments, stretching modulus zone **418** of first upper portion **414** may be different from stretching modulus zone **420** of first lower portion **416** such that they have different levels of stretchability. In other embodiments, stretching modulus zone **418** of first upper portion **414** may have the same level of stretchability as stretching modulus zone **420** first lower portion **416**.

[0042] Similarly, on second side portion **406**, a second upper portion **422** extends from collar **410** in a downward direction towards bottom hem **412**. Second upper portion **422** is associated with a stretching modulus zone **426** having a level of stretchability. Second side portion **406** also includes a second lower portion **424** that extends in an upward direction from bottom hem **412** towards collar **410**. Second lower portion **424** is associated with a stretching modulus zone **428** having a level of stretchability. In some embodiments, stretching modulus zone **426** of second upper portion **422** may be different from stretching modulus zone **428** of second lower portion **424** such that they have different levels of stretchability. In other embodiments, stretching modulus zone **426** of second upper portion **422** may have the same level of stretchability as stretching modulus zone **428** of second lower portion **424**.

[0043] Additionally, in some embodiments, the levels of stretchability may be the same across similar parts or sections of article **400** such that stretching modulus zone **418** of first upper portion **414** on first side portion **404** may have the same level of stretchability as stretching modulus zone **426** of second upper portion **422** on second side portion **406**. Similarly, stretching modulus zone **420** of first lower portion **416** on first side portion **404** may have the same level of stretchability as stretching modulus zone **428** of second lower portion **424** on second side portion **406**.

[0044] In other embodiments, the levels of stretchability may be different across similar parts or sections of article **400** such that stretching modulus zone **418** of first upper portion **414** on first side portion **404** is different from stretching modulus zone **426** of second upper portion **422** on second side portion **406** such that they have different levels of stretchability. Similarly, stretching modulus zone **420** of first lower portion **416** on first side portion **404** may be different from stretching modulus zone **428** of second lower portion **424** on second side portion **406** such that they have different levels of stretchability.

[0045] The single-layer woven portion of the engineered woven fabric forming article **400** also includes parts or sections immediately surrounding each of first pocket **110** on first side portion **404** and second pocket **112** on second side portion **406**. As shown in FIG. **4**, a first surrounding portion **430** is disposed around the edges of first pocket **110** on first side portion **404** and a second surrounding portion **434** is disposed around the edges of second pocket **112** on second side portion **406**. In an example embodiment, first surrounding portion **430** is associated with a stretching modulus zone **432** having a level of stretchability and second surrounding portion **434** is associated with a stretching modulus zone **436** having a level of stretchability.

[0046] In an example embodiment, stretching modulus zone **432** of first surrounding portion **430** may have the same level of stretchability as stretching modulus zone **436** of second surrounding portion **434**. In other embodiments, however, stretching modulus zone **432** of first surrounding portion **430** on first side portion **404** may be different from stretching modulus zone **436** of second surrounding portion **434** on second side portion **406** such that they have different levels of stretchability.

[0047] In accordance with the techniques described herein, one or more of the stretching modulus zones in different parts or sections of the engineered woven fabric of article of clothing **400** may have different levels of stretchability. In an example embodiment, one or more of stretching modulus zone **418** of first upper portion **414**, stretching modulus zone **420** of first lower portion **416**, stretching modulus zone **426** of second upper portion **422**, and stretching modulus zone **428** of second lower portion **424** may be a first stretching modulus zone with a first level of stretchability. One or more of stretching modulus zone **432** of first surrounding portion **430** and stretching modulus zone **436** of second surrounding portion **434** may be a second stretching modulus zone with a second level of stretchability. In some embodiments, the first level of stretchability is greater than the second level of stretchability such that the parts or sections of article **400** corresponding to the first stretching modulus zone(s) stretch to a greater amount or degree than the parts or sections of article **400** corresponding to the second stretching modulus zone(s).

[0048] With this arrangement, the second stretching modulus zones having lesser levels of stretchability are disposed around the edges of the pockets in article **400** (e.g., first surrounding portion **430** extending around the edges of first pocket **110** and second surrounding portion **434** extending around the edges of second pocket **112**) to provide support to the pockets and resist stretching to the same amount or degree as the remaining parts or sections of article **400** (e.g., first upper portion **414**, first lower portion **416**, second upper portion **422**, and/or second lower portion **424**) that are associated with the first stretching modulus zones having greater levels of stretchability. By providing a lesser amount or degree of stretch to the parts or sections of article **400** immediately surrounding the pockets **110**, **112**, article **400** may have a smoother overall appearance and may resist or prevent bagging or sagging of pockets **110**, **112** when carrying items or objects.

[0049] In some embodiments, an entirety of the single-layer woven portion of the engineered woven fabric forming article **400** may have a level of stretchability that is less than a level of stretchability of the double-layer woven portions that form each of first pocket **110** and second pocket **112**. With this arrangement, the double-layer woven portions forming each of first pocket **110** and second pocket **112** may stretch to a greater degree than the single-layer woven portions of the engineered woven fabric forming article **400** to allow or assist with opening of first pocket **110** and second pocket **112**.

[0050] Additionally, in some embodiments, a level of stretchability of each layer of the double-layer woven portions forming first pocket **110** and second pocket **112** may be varied. For example, a first layer of the double-layer woven portion may form a front layer of the pocket (e.g., associated with an exterior of article **400**) and a second layer may form the opposite back layer of the pocket (e.g., associated with an interior of article **400**) for each of first pocket **110** and second pocket **112**. In an example embodiment, each of these front and back layers may have different levels of stretchability.

[0051] In one embodiment, the level of stretchability of the front and back layers of the pocket may be varied by changing a yarn density of each layer during the weaving process forming article **400**. For example, when one quarter (¼) of the total number of yarns forming article **400** are used to form the front layer and the remaining three quarters (¾) of the total number of yarns are used to form the back layer, this variation in yarn density between the front layer and the back layer of the pocket will allow the front layer to stretch to a greater degree than the back layer. Similar techniques may be used to vary the stretchability of one or more double-layer woven portions of article **400** that may be located in different areas or extend through a larger portion of article **400** than first pocket **110** and second pocket **112**.

[0052] In some embodiments, the second stretching modulus zone may be woven as a no-stretch zone so as to substantially prevent stretching. In other embodiments, the second stretching modulus zone may be woven using a reverse twill weave pattern or technique. In still other embodiments, the second stretching modulus zone may be woven using a plain weave pattern or technique. In some embodiments, the first stretching modulus zone may be woven using a regular twill weave pattern or technique. In other embodiments, the first stretching modulus zone may be woven using a ripstop weave pattern or techniques. In still other embodiments, the first stretching modulus zone may be woven using an elastane yarn. For example, the elastane yarn may be added in the warp direction and/or or the weft direction of the parts or sections of article **400** associated with the first stretching modulus zone.

[0053] Additionally, in some embodiments, each independent woven layer of the double-layer woven portion (e.g., first woven layer **302** and second woven layer **304** of first pocket **110** shown in FIG. **3**) may also be formed with different stretching modulus zones. For example, in one embodiment, one woven layer of the double-layer woven portion may have a stretching modulus zone that is different from a stretching modulus zone of the other woven layer of the double-layer woven portion. In some cases, one of the stretching modulus zones of the double-layer woven portion may have a greater level of stretchability than the opposite layer. That is, one woven layer of the double-layer woven portion may stretch to a larger amount or degree than the other woven layer. In one embodiment, one woven layer may be made as a no-stretch modulus zone (i.e., to substantially prevent stretching) while the other woven layer may be made to have some amount or degree of stretch, for example, by incorporating an elastane yarn or other suitable material exhibiting some amount or percentage of stretch.

[0054] Referring now to FIG. **5**, an enlarged schematic view of an example embodiment of pocket **110** formed in the engineered woven fabric of article **400** is shown surrounded by different stretching modulus zones. In an example embodiment, first pocket **110** includes outer perimeter **200** defined by four edges, including first edge **202**, second edge **204**, third edge **206**, and fourth edge **208**, as described above. In this embodiment, three of the four edges of first pocket **110** are closed edges (i.e., first edge **202**, second edge **204**, and third edge **206**) that are seamlessly joined with the surrounding woven fabric of article **400**. The double-layer woven portion forming first pocket **110** includes a remaining edge (i.e., fourth edge **208**) that has one woven layer that is open (i.e., free edge **210**) to provide opening **114** into the cavity or void space of first pocket **110**.

[0055] In one embodiment, the double-layer woven portion forming first pocket **110** is formed by weaving using a first weave pattern **500**. In an example embodiment, first weave pattern **500** may be a plain weave pattern or technique. In other embodiments, first weave pattern **500** may another type of weaving pattern or technique. This double-layer woven first pocket **110** formed using first weave pattern **500** is surrounded on at least three sides by a single-layer woven portion forming first surrounding portion **430**. In an example embodiment, first surrounding portion **430** includes an outer perimeter that is spaced apart from outer perimeter **200** of first pocket **110** and includes a first edge **502**, a second edge **504**, and a third edge **506**. As shown in FIG. **5**, first surrounding portion **430** extends around or surrounds the closed edges of first pocket **110**, including first edge **202**, second edge **204**, and third edge **206**.

[0056] In some embodiments, the size or area of first surrounding portion **430** surrounding the closed edges of first pocket **110** may vary based on the size or area of first pocket **110**. In some cases, the size or area of first surrounding portion **430** may be a predetermined percentage of the size or area of first pocket **110**. For example, in one embodiment, the size or area of first surrounding portion **430** may be approximately 10% of the size or area of first pocket **110** (e.g., for a pocket having an area of 250 square centimeters, the surrounding portion may be at least 25 square centimeters). In other embodiments, the predetermined percentage may be smaller or larger. For example, in some cases, between 5%-25% of the size or area of the pocket. Additionally, in some cases, the predetermined percentage may vary based on the level of stretchability of the pocket and/or the surrounding portion around the pocket. For example, a surrounding portion formed using a weave that creates a no-stretch zone may be a smaller predetermined percentage of the size or area of the pocket compared with a surrounding portion formed using a weave that has a non-zero level of stretchability.

[0057] In one embodiment, the single-layer woven portion forming first surrounding portion **430** is formed by weaving using a second weave pattern **508**. In one embodiment, second weave pattern **508** may be a reverse twill or herringbone weave pattern or technique. In other embodiments, second weave pattern **508** may be a plain weave, a type of weave forming a no-stretch zone, or other types of weaving patterns or techniques. In an example embodiment, second weave pattern **508** is different from first weave pattern **500**. As described above, in some embodiments, first surrounding portion **430** having second weave pattern **508** may be associated with stretching modulus zone **432** having a level of stretchability (e.g., a second level of stretchability that is less than a first level of stretchability) that is less than a stretching modulus zone of at least one other part or portion of the engineered woven fabric.

[0058] For example, in this embodiment, first surrounding portion **430** is itself partially surrounded by parts or sections of first upper portion **414** and/or first lower portion **416**. As shown in FIG. **5**, a top edge **510** of first upper portion **414** surrounds at least first edge **502** of first surrounding portion **430**, a side edge **512** of first upper portion **414** and/or first lower portion **416** surrounds at least second edge **504** of first surrounding portion **430**, and a bottom edge **514** of first lower portion **416** surrounds at least third edge **506** of first surrounding portion **430**.

[0059] In one embodiment, the single-layer woven portion forming first upper portion **414** and/or first lower portion **416** is formed by weaving using a third weave pattern **516**. In one embodiment, third weave pattern **516** may be a regular twill weave pattern or technique. In other embodiments, third weave pattern **516** may be a ripstop weave, a type of weave including at least one elastane yarn, or other types of weaving patterns or techniques. In an example embodiment, third weave pattern **516** is different from second weave pattern **508** and/or first weave pattern **500**. As described above, in some embodiments, first upper portion **414** and/or first lower portion **416** having third weave pattern **516** may be associated with stretching modulus zone **418**, **420** having a level of stretchability that is greater than a stretching modulus zone of first surrounding portion **430** (e.g., a first level of stretchability that is more than a second level of stretchability).

[0060] In some embodiments, the level of stretchability of stretching modulus zone **418**, **420** in first upper portion **414** and/or first lower portion **416** having third weave pattern **516** is smaller than a level of stretchability of first pocket **110**. With this arrangement, a gradient in the level of stretchability may be formed starting at the area of article **400** at opening **114** of first pocket **110**, where the level of stretchability may be small or exhibit no stretch properties, and may increase to permit or allow more stretch as a distance from opening **114** increases. For example, as shown in FIG. **4**, stretching modulus zone **432** in first surrounding portion **430** is located closer to opening **114** and exhibits less stretch than stretching modulus zone **418**, **420** in first upper portion **414** and/or first lower portion **416** which are located farther away from opening **114**.

[0061] It should be understood that a substantially similar arrangement may be provided on second side portion **406** to surround second pocket **112** with parts or sections of the engineered woven fabric having different stretching modulus zones as described above with regard to first pocket **110**. With this arrangement, portions of article **400** having a lesser amount or degree of stretch, for example, first surrounding portion **430** and/or second surrounding portion **434** (shown in FIG. **4**), may be located immediately surrounding pockets **110**, **112**, and the remaining parts or sections of article **400** disposed around first surrounding portion **430** and/or second surrounding portion **434** may have a greater amount or degree of stretch, for example, first upper portion **414**, first lower portion **416**, second upper portion **422**, and/or second lower portion **424** to provide article **400** with a smoother overall appearance and to resist or prevent bagging or sagging of pockets **110**, **112** when carrying items or objects.

[0062] The principles of the present embodiments have been described herein with reference to a front side of an article (e.g., front side **102** of article **100** and/or front side **402** of article **400**), however, the techniques may also be applied to other parts or sections of an article, including but not limited to a back side and/or side panels. For example, an article may include one or more parts or sections on a back side and/or side panels that are woven using a rip-stop weave or that include elastane yarn to provide varying levels of stretchability in a similar manner as described with reference to the example embodiments herein. Additionally, the location and/or arrangement of pockets and stretching modulus zones may vary based on the type of article or garment.

[0063] In some embodiments, articles of clothing (e.g., article **100** and/or article **400**) described herein may be seamlessly woven during the same weaving process, including one or more parts or sections having a single-layer woven construction and one or more parts or sections having a double-layer woven construction (e.g., pockets **110**, **112**). In an example embodiment, after the weaving process forming the engineered woven fabric is completed, additional steps or processes may be applied to the engineered woven fabric to fashion or form it into a specific article of clothing, including article **100** and/or article **400**. In one embodiment, the pockets (e.g., pockets **110**, **112**) may be initially formed during the weaving process of the engineered woven fabric with all edges around the outer perimeter closed (i.e., seamlessly joined with the surrounding fabric). In such cases, a step of creating or forming the openings in pockets **110**, **112** may be performed upon completion of the weaving process to provide access or entry into the cavity of void space formed between the independent woven layers of pockets **110**, **112**.

[0064] Referring now to FIG. **6**, an enlarged representative view of a process **600** for seamlessly forming an opening in a pocket in an engineered woven fabric is shown. In this embodiment, the engineered woven fabric includes a single-layer woven portion **602** that is seamlessly joined with a double-layer woven portion **604**. As described above, in some embodiments, the cavity or void space between the independent woven layers of double-layer woven portion **604** may be fashioned into a pocket by opening one of the closed edges. In an example embodiment, double-layer woven portion **604** has an outer perimeter **606** delimiting the boundaries of a pocket defined by four edges, including a first edge **608**, a second edge **610**, a third edge **612**, and a fourth edge **614**.

[0065] In one embodiment, one of the closed edges around outer perimeter **606** of double-layer woven portion **604** may be opened to provide an opening and form the pocket. In this embodiment, process **600** includes using a cutting device **616**, such as a knife, razor, or other cutting mechanism, to cut through one or more yarns of double-layer woven portion **604** along fourth edge **614**. As shown in FIG. **6**, cutting device **616** may be moved in a downward direction **618** (e.g., from first edge **606** towards third edge **612**) to open double-layer woven portion **604** along fourth edge **614**.

[0066] FIG. **7**A shows double-layer woven portion **604** seamlessly joined to single-layer woven portion **602** at both ends with a cavity or void space **700** formed between the independent woven layers of double-layer woven portion **604**, including a first woven layer **702** and a second woven layer **704**. As shown in FIG. **7**B, process **600** is implemented by using cutting device **616** to cut through yarns of first woven layer **702** of double-layer woven portion **604** (e.g., the top layer in this embodiment) to separate them from single-layer woven portion **602**. As shown in FIG. **7**C, once yarns of first woven layer **702** of double-layer woven portion **604** are cut or detached from single-layer woven portion **602**, the cut portion of double-layer woven portion **604** becomes a free edge **706** that may be moved away from second woven layer **704** to provide an opening **708** that allows access into cavity or void space **700** located between the two independent woven layers **702**, **704** of double-layer woven portion **604**. With this arrangement, a pocket may be formed in the engineered woven fabric.

[0067] It should be understood that process **600** is one example of a method for forming a pocket in an engineered woven fabric. Other techniques may be used, including other mechanisms or techniques for cutting open one edge of the double-layer woven portion to form the pocket, as well as integrally forming a free end as part of the double-layer woven portion during the weaving process itself.

[0068] The present embodiments have described herein in reference to an exemplary pocket having a generally rectangular or square shape. In some embodiments, a pocket formed according to the techniques described herein may have other shapes, including shapes having fewer or greater number of sides or edges. Referring now to FIGS. **8**A-**8**C, some additional embodiments of pockets having non-rectangular shapes having varying numbers of sides or edges are shown. For example, FIG. **8**A illustrates an alternate embodiment of an arrangement for a pocket **800** formed in the engineered woven fabric of an article. In this embodiment, pocket **800** includes a first straight edge **802**, a curved bottom edge **804**, and a second straight edge **806** that may be seamlessly joined with an engineered woven fabric. Pocket **800** also includes a free edge **808** disposed along the top of pocket **800** that provides an opening into pocket **800**.

[0069] FIG. **8**B illustrates another alternate embodiment of an arrangement for a pocket **810** formed in the engineered woven fabric of an article. In this embodiment, pocket **810** includes a first straight edge **812**, a first angled bottom edge **814**, a second angled bottom edge **816**, and a second straight edge **818** that may be seamlessly joined with an engineered woven fabric. Pocket **810** also includes a free edge **820** disposed along the top of pocket **810** that provides an opening into pocket **810**. FIG. **8**C is another alternate embodiment of an arrangement for a pocket **822** formed in the engineered woven fabric of an article. In this embodiment, pocket **822** includes a first edge **824** and a second edge **826** that may be seamlessly joined with an engineered woven fabric. Pocket **822** also includes a free edge **828** disposed along the top of pocket **822** that provides an opening into pocket **822**.

[0070] Additionally, while the present embodiments have been described in reference to an exemplary embodiment of an article of clothing in the form of a vest (e.g., article **100** and/or article **400**), the techniques described herein may be applied to a variety of different types and forms of clothing or garments. For example, FIGS. **9**A-**9**E illustrate different example embodiments of articles of clothing including an engineered woven fabric with pockets formed according to the techniques described herein. Referring now to FIG. **9**A, an article of clothing **900** in the form of a coat or jacket is shown. In this embodiment, article **900** may include one or pockets formed in accordance with the techniques described herein, including, but not limited to a first side pocket **902**, a second side pocket **904**, a first chest pocket **906**, and a second chest pocket **908**.

[0071] Referring now to FIG. **9**B, an article of clothing **910** in the form of a shirt or t-shirt is shown. In this embodiment, article **910** may include one or pockets formed in accordance with the techniques described herein, including, but not limited to a chest pocket **912**. In some embodiments, articles of clothing that may be worn over the legs or lower body of a wearer may also have pockets formed in the engineered woven fabric as described above. For example, FIG. **9**C illustrates an article of clothing **920** in the form of a pair of pants. In this embodiment, article **920** may include one or pockets formed in accordance with the techniques described herein, including, but not limited to a first pocket **922** and a second pocket **924**. Additionally, in some cases, article **920** may also include pockets on the back side (not shown).

[0072] Referring now to FIG. **9**D, an article of clothing **930** in the form of a pair of shorts is shown. In this embodiment, article **930** may include one or pockets formed in accordance with the techniques described herein, including, but not limited to a first pocket **932** and a second pocket **934**. Additionally, in some cases, article **930** may also include pockets on the back side (not shown). Referring now to FIG. **9**E, an article of clothing **940** in the form of a skirt or dress is shown. In this embodiment, article **940** may include one or pockets formed in accordance with the techniques described herein, including, but not limited to a first pocket **942** and a second pocket **944**.

[0073] The example embodiments of articles of clothing or garments shown in FIGS. **9**A-**9**E should not be considered limiting. Other types or forms of articles of clothing or garments may be formed with pockets made according to the example embodiments as would be obvious to one of ordinary skill in the art in view of the present teachings.

[0074] While various embodiments of the disclosure have been described, the description is intended to be exemplary, rather than limiting and it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the disclosure. Accordingly, the disclosure is not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

Claims

1. An article of clothing including an engineered woven fabric comprising: a single-layer woven portion having at least one warp yarn and at least one weft yarn, the single-layer woven portion engineered with a first stretching modulus zone with a first level of stretchability and a second stretching modulus zone with a second level of stretchability, the first level of stretchability being greater than the second level of stretchability; and a double-layer woven portion seamlessly joined with the single layer woven portion having two independent woven layers each having at least one warp yarn and at least one weft yarn, the double-layer woven portion forming a pocket with closed edges seamlessly joined with the single-layer woven portion and an opening; wherein the single-layer woven portion is engineered such that the second stretching modulus zone is disposed around the closed edges of the pocket to provide support around the pocket.

2. The article of clothing of claim 1, wherein the second stretching modulus zone is a no-stretch zone, a reverse twill weave, or a plain weave.

3-4. (canceled)

5. The article of clothing of claim 1, wherein the first stretching modulus zone comprises an elastane yarn, wherein the elastane yarn is added in the warp direction and/or the weft direction of the first stretching modulus zone.

6. (canceled)

7. The article of clothing of claim 1, wherein the first stretching modulus zone is a ripstop weave.

8. The article of clothing of claim 1, wherein the single-layer woven portion comprises at least one additional stretching modulus zone with a level of stretchability different from the first and the second level of stretchability.

9. The article of clothing of claim 1, wherein the single-layer woven portion comprises two or more warp yarns and two or more weft yarns, and/or wherein the double-layer woven portion comprises two or more warp yarns and two or more weft yarns.

10. The article of clothing of claim 1, wherein the double-layer woven portion is engineered with a third stretching modulus zone with a third level of stretchability and a fourth stretching module zone with a fourth level of stretchability, the third level of stretchability being greater than the fourth level of stretchability.

11. The article of clothing of claim 10, wherein the fourth stretching modulus zone is a no-stretch modulus zone.

12. The article of clothing of claim 10, wherein the third stretching modulus zone comprises an elastane yarn, wherein the elastane yarn is added in the warp direction and/or the weft direction of the third stretching modulus zone.

13. (canceled)

14. The article of clothing of claim 1, further comprising a gradient in the level of stretchability of the single-layer woven portion that increases in stretchability as a distance from the opening of the pocket increases.

15. The article of clothing of claim 14, wherein the second stretching modulus zone is a no-stretch zone disposed adjacent to the opening of the pocket; wherein the first stretching modulus zone is located a distance away from the opening of the pocket; and wherein the gradient in the level of stretchability increases from the second level of stretchability in the second stretching modulus zone to the first level of stretchability in the first stretching modulus zone.

16-17. (canceled)

18. The article of clothing of claim 1, wherein one woven layer of the two independent woven layers forming the double-layer woven portion has a level of stretchability that is greater than the other woven layer.

19. An article of clothing including an engineered woven fabric comprising: a pocket formed in the engineered woven fabric, the pocket comprising a plurality of closed edges disposed around an outer perimeter of the pocket and a free edge that provides an opening into a cavity of the pocket; a surrounding portion of the engineered woven fabric surrounding at least two closed edges of the plurality of closed edges around the outer perimeter of the pocket; and at least one additional portion of the engineered woven fabric surrounding the surrounding portion; wherein the at least one additional portion of the engineered woven fabric includes a first stretching modulus zone having a first level of stretchability; wherein the surrounding portion includes a second stretching modulus zone having a second level of stretchability, the second level of stretchability being less than the first level of stretchability to provide support around the pocket.

20. The article of clothing of claim 19, wherein the surrounding portion comprises a single-layer woven portion having at least one warp yarn and at least one weft yarn; and wherein the pocket comprises a double-layer woven portion having two independent woven layers each having at least one warp yarn and at least one weft yarn, the double-layer woven portion of the pocket being seamlessly joined with the single layer woven portion of the surrounding portion along each of the closed edges.

21. The article of clothing of claim 20, wherein one woven layer of the two independent woven layers is detached from the other woven layer to form the free edge.

22. The article of clothing of claim 20, wherein one woven layer of the two independent woven layers has a level of stretchability that is greater than the other woven layer.

23. The article of clothing of claim 22, wherein the one woven layer of the two independent woven layers having the greater level of stretchability comprises one quarter of a total number of yarns forming the engineered woven fabric and the other woven layer comprises three quarters of the total number of yarns forming the engineered woven fabric.

24. The article of clothing of claim 19, wherein the first stretching modulus zone is associated with a first weave pattern and the second stretching modulus zone is associated with a second weave pattern that is different from the first weave pattern.

25. The article of clothing of claim 19, wherein the engineered woven fabric has a gradient in the level of stretchability that increases in stretchability as a distance from the opening of the pocket increases.

26. A method of forming an article of clothing including an engineered woven fabric having a seamless pocket, the method comprising: weaving an engineered woven fabric comprising: (1) a single-layer woven portion having at least one warp yarn and at least one weft yarn, and (2) a double-layer woven portion having two independent woven layers each having at least one warp yarn and at least one weft yarn, wherein the double-layer woven portion is seamlessly joined with the single layer woven portion during the weaving process; wherein weaving the single-layer woven portion further comprises: weaving a first stretching modulus zone in a first portion of the engineered woven fabric with a first level of stretchability; and weaving a second stretching modulus zone in a second portion of the engineered woven fabric with a second level of stretchability, the first level of stretchability being greater than the second level of stretchability; wherein weaving the double-layer woven portion further comprises forming a pocket in the engineered woven fabric with closed edges extending around an outer perimeter of the pocket, the closed edges being seamlessly joined with the single-layer woven portion; and cutting one woven layer of the two independent woven layers along one edge of the closed edges extending around the outer perimeter of the pocket to detach the one woven layer and form a free edge, wherein the free edge provides an opening into a cavity formed between the two independent woven layers of the pocket.